

CaffeinatedData

Linear Regression Project

Research Question:

What customer and service-related factors are associated with changes in **MonthlyCharge** within this organization?

Business Goal:

The goal of this analysis is to develop a multiple linear regression model that identifies which factors relating to customer and service-related characteristics significantly influence **MonthlyCharge**. By quantifying the relationship between customer characteristics, demographic attributes and service features, the company can better understand the drivers of pricing variability. These insights can be used to improve revenue forecasting, support pricing strategy decisions and inform efforts at improving customer retention.

Data Analysis Process:

1. There should be a linear relationship between the dependent variable and each independent variable. This means that changes in the independent variable are associated with proportional changes in the dependent variable. For example, **MonthlyCharge** should change roughly linearly with **Bandwidth_GB_Year**. Linearity implies that as X(independent) increases or decreases, Y(dependent) changes accordingly to match proportionally in a predictable manner.
2. The independent variables should not be too highly correlated with each other in multiple linear regression. When the explanatory variables are strongly correlated, the model cannot accurately distinguish which variable to be mostly correlated with the response variable. This issue, **multicollinearity**, can reduce reliability and accuracy of the coefficient estimates.
3. Observations(rows) should be independently and randomly selected from the population. Each data point should represent a separate and unrelated observation, such as that one observation does not influence another. In this dataset, each row represents an individual customer from across the country, independent of one another, satisfying this assumption.
4. The residuals of multiple linear regression models should be approximately normally distributed (bell shape) with a mean of zero (center of bell is at 0) and show constant variance across all predicted values. While constant variance shows that the model's errors are evenly distributed and do not increase or decrease with predicted values. Residuals represent the difference between actual and predicted points.

Justification for the chosen tools:

Python provides powerful libraries that support efficient data cleaning and preparation. Tools such as Pandas and Numpy makes it possible to create data frames, manipulate data, handle missing values and transform data into a suitable format for linear regression. These capabilities improve data quality and ensure that the analysis is solid.

Python also offers support for statistical modeling and visualization. Libraries such as matplotlib and seaborn allows for visualization of the data and evaluate the data. Additionally, the stats model package provides an easy way to create a linear regression model through its implementation of Ordinary Least Squares regression and generate detailed model summaries through its **summary()** method, allowing easier interpretation of coefficients and model performance.

Justification for chosen model:

Multiple linear regression is an appropriate technique for this analysis because the research question examines how multiple independent variables can influence a continuous dependent variable **MonthlyCharge**. The model allows the effects of multiple independent variables to be evaluated while controlling the influence of other predictors. This approach allows for quantifying the individual contribution of each variable and identify key drivers of pricing and support data-driven decision making.

Data Cleaning Process:

The primary goal of data cleaning is to provide a high-quality data set with relevant variables for multiple linear regression involving **MonthlyCharge** as the response variable. This included ensuring completeness, consistency and appropriate data types for all dependent and independent variables used in the model.

The dataset was reduced to include only variables relevant to the research question, focusing on customer demographics, service characteristics and usage metrics believed to influence **MonthlyCharge**. This step minimizes noise and improved model interpretability.

The data set is loaded through **pd.read_csv** with **na_values** argument to catch null values right at the beginning

```
#Importing Data and removing nulls and nans
```

```
filename = 'churn_clean.csv'
```

```
df = pd.read_csv(filename, na_values = ['null','N/A',' ', ' ','.', 'nan'])
```

```
#Reducing the Dataset to relevant columns for multiple Linear Regression
```

```
df_model = df[
```

```
[
```

```
    'MonthlyCharge',
```

```
    'Bandwidth_GB_Year',
```

```
    'Income',
```

```
    'Tenure',
```

```
    'Population',
```

```
    'Children',
```

```
    'Contract',
```

```
    'InternetService',
```

```
    'Phone',
```

```
'StreamingTV',  
'StreamingMovies',  
'TechSupport',  
'OnlineSecurity',  
'Area',  
  
]  
]
```

Missing values were assessed across all selected variables to identify incomplete data that could affect model estimation.

#Checking for missing values across all columns in the subsetted dataset

```
print(df_model.isna().sum())
```

The InternetService variable contained 2,129 missing values. Examination of category frequencies suggested that these missing values likely represented customers without internet service from the provider. To preserve this information and avoid misclassification, missing values were replaced with a meaningful category labeled "None."

```
print(df_model['InternetService'].value_counts())
```

#Filling InternetService missing data with "None", since the customer could have internet service from another company.

#The customer possibly only uses phone/tv services from this company

```
df_model['InternetService'] = df_model['InternetService'].fillna('None')
```

```
print(df_model['InternetService'].value_counts())
```

#Checking subsetted dataset

```
df_model.isnull().sum()
```

#Checking Datatypes of subset dataset

```
df_model.dtypes
```

Data types were then reviewed to ensure compatibility with regression analysis. Continuous variables were retained as numeric types, and categorical variables were explicitly converted to categorical data types.

```
# Convert categorical variables to category dtype for regression analysis
```

```
cat_cols = [  
    'Contract',  
    'InternetService',  
    'Phone',  
    'StreamingTV',  
    'StreamingMovies',  
    'TechSupport',  
    'OnlineSecurity',  
    'Area'  
]
```

```
for col in cat_cols:
```

```
    df_model[col] = df_model[col].astype('category')
```

```
#Double check dtypes
```

```
df_model.dtypes
```

Potential outliers in continuous variables were assessed using boxplots.

#Creating Boxplot with Income, to check for outliers

```
income_box = sns.boxplot(x='Income',data=df_model)
```

#Boxplot for MonthlyCharge, to check for outliers

```
sns.boxplot(x='MonthlyCharge',data=df_model)
```

#Boxplot for Bandwidth_GB_Year, to check for outliers

```
sns.boxplot(x='Bandwidth_GB_Year',data=df_model)
```

#Boxplot for Tenure, to check for outliers

```
sns.boxplot(x='Tenure',data=df_model)
```

#Boxplot for Population, to check for outliers

```
sns.boxplot(x='Population',data=df_model)
```

#Boxplot for Children, to check for outliers

```
sns.boxplot(x='Children',data=df_model)
```

The Income variable exhibited significant right skew due to a small number of extreme values. To reduce the influence of outliers while preserving all observations, income values above \$175,000 were capped at \$175,000.

```
#Capping max values at 175k
```

```
df_model.loc[df_model['Income'] > 175000, 'Income'] = 175000
```

```
#Checking max income and reassurance
```

```
print(df_model['Income'].max())
```

```
print(df_model['Income'].count())
```

While variables Children and Population also carries outliers, both variables kept their outliers because Population's outliers are reflections of larger urban area's and Children's outliers could reflect higher MonthlyCharges.

```
#Boxplot for Children, to check for outliers
```

```
sns.boxplot(x='Children',data=df_model)
```

```
print((df_model['Children'] > 7).sum())
```

Describe the Variables:

Dependent Variable: MonthlyCharge

MonthlyCharge is a continuous numerical variable representing the amount in dollars for the customer's monthly bill. The mean of MonthlyCharge is approximately \$172.62, with a median of \$167.48. MonthlyCharge values range from \$79.97 to \$290.16, resulting in a range of \$210.

The standard deviation of \$42.94 suggest moderate variability in monthly pricing across customers, reflecting difference in service usages, additional choices in plans and features.

First quartile is \$139.97 and third quartile is \$200.73. The distribution is unimodal.

```
count    10000.000000
mean      172.624816
std       42.943094
min       79.978860
25%      139.979239
50%      167.484700
75%      200.734725
max       290.160419
Name: MonthlyCharge, dtype: float64
Range: 210.181559
```

Independent Variables:

Bandwidth_GB_Year:

Bandwidth_GB_Year is a continuous numerical variable representing a customer's annual data usage. The mean of annual bandwidth usage is approximately 3,392 gb, with a median of 3,280 gb, indicating a relatively symmetric distribution. Bandwidth usage ranges from approximately 156 gb to 7,159 gb, resulting in a range of about 7,003 gb. The standard deviation of 2185 suggest variability in data usage among customers, which signifies it could be a strong potential predictor of MonthlyCharge. The first quartile is 1236.47gb and the third quartile is 5586.14 gb. The distribution is bimodal with two distinct peaks, suggesting two customer groups with differing levels of bandwidth usage.

```
count    10000.000000
mean      3392.341550
std       2185.294852
min       155.506715
25%      1236.470827
50%      3279.536903
75%      5586.141370
max       7158.981530
Name: Bandwidth_GB_Year, dtype: float64
Range: 7003.4748152
```

Income:

Income is a continuous numerical variable representing the customers' annual income as reported at the time of sign-up. The mean of Income is approximately \$39,768, with a median of \$33,170, suggesting a slight right skew. Reported Income ranges from \$348 to a capped maximum of \$175,000 following data cleaning. Several extreme high-income values were observed beyond the upper whiskers of the boxplot; however, these observations became sparse beyond \$175,000. To reduce the influence of extreme outliers, income values above \$175,000 were capped at this threshold. The standard deviation of Income is approximately \$27,976. The first quartile of the distribution is \$19,224, the third quartile is \$53,246. The distribution is a unimodal, with a strong right skew due to reported incomes at \$175,000.

```
count    10000.000000
mean      39768.591427
std       27976.727587
min        348.670000
25%       19224.717500
50%       33170.605000
75%       53246.170000
max      175000.000000
Name: Income, dtype: float64
Range: 174651.33
```

Tenure:

Tenure is a continuous numerical variable representing the number of months the customer has stayed with this service provider. The mean of Tenure is approximately 34.52 months, with a median of 35.43 months. The lowest Tenure was 1 month while the highest tenure is 71.99 months, with a range of 70.99 months. The standard deviation of Tenure is 26.44 months.

The distributions first quartile is 7.91 months, the third quartile at 61.47 months. The distribution is bimodal, suggesting two customer groups with differing tenures.

```
count    10000.000000
mean      34.526188
std       26.443063
min        1.000259
25%        7.917694
50%       35.430507
75%       61.479795
max       71.999280
Name: Tenure, dtype: float64
Range: 70.99902066
```

Population:

Population is a continuous numerical variable representing the population within a mile radius of the customer. The mean of Population is 9756.56, with a median of 2910.5. The lowest record of Population is 0 while the largest record is 111850, with a range of 111850. The standard deviation of Population is 14432.69. The distributions first quartile is 738, the third quartile is 13168. The shape of the distribution is unimodal with a strong right skew due to high population density, suggesting urban metropolitan cities pull the mean up higher as the mean is much larger than the median.

```
count    10000.000000
mean      9756.562400
std      14432.698671
min         0.000000
25%       738.000000
50%      2910.500000
75%     13168.000000
max     111850.000000
Name: Population, dtype: float64
Range: 111850
```

Children:

Children is a continuous numerical variable representing the number of children in the customers household at the time of sign up. The mean of Children is 2.08, with a median of 1. The lowest record for Children is 0 and the largest is 10. Children has a standard deviation of 2.14, with the first quartile at 0 and third quartile at 3. The distribution is shaped as a unimodal with a strong right skew due to outliers.

```
count    10000.0000
mean       2.0877
std       2.1472
min        0.0000
25%        0.0000
50%        1.0000
75%        3.0000
max       10.0000
Name: Children, dtype: float64
Range: 10
```

Contract:

Contract is a categorical ordinal variable representing the customer's billing agreement length. The contract categories follow an order from shortest to longest of "Month-to-month", "One Year", "Two Year" contracts. Month-to-month contracts are the most common, accounting at 54.4%, followed by Two Year at 24.4%, then One Year at 21%. Contract length is relevant to MonthlyCharge because longer term contracts often include pricing incentives or discounts.

```

Contract
Month-to-month    5456
Two Year          2442
One year          2102
Name: count, dtype: int64
Contract
Month-to-month    54.56
Two Year          24.42
One year          21.02
Name: proportion, dtype: float64

```

Phone:

Phone is a categorical binary variable representing if customers signed up for phone service through the company. The column shows most customers do have phone service through the company at 9067 records, a percentage of 90.67% while those who do not are at 933 records, at 9.33%. This variable is relevant to MonthlyCharge as customers could be paying for a phone service and internet service, compounding to their monthly charges.

```

Phone
Yes    9067
No      933
Name: count, dtype: int64
Phone
Yes    90.67
No      9.33
Name: proportion, dtype: float64

```

InternetService:

InternetService is a categorical nominal variable representing the type of internet service used by the customer. The categories include Fiber Optic, DSL and None. Fiber Optic is the most common observation at 44%, followed by DSL at 34.63%, then lastly at None at 21.29%. While a smaller portion of customers do not subscribe to internet service through this provider, internet service type is relevant to MonthlyCharge as faster internet commands a higher price.

```

InternetService
Fiber Optic    4408
DSL            3463
None           2129
Name: count, dtype: int64
InternetService
Fiber Optic    44.08
DSL            34.63
None           21.29
Name: proportion, dtype: float64

```

StreamingTV:

StreamingTV is a binary categorical variable representing customers who opted for streaming television services through the company. The dataset shows most customers do not subscribe to streaming television services at 50.71% while those who opted for it are at 49.29%. This variable is relevant to MonthlyCharge because customers who do subscribe to additional entertainment typically incur higher monthly cost.

```
StreamingTV
No      5071
Yes     4929
Name: count, dtype: int64
StreamingTV
No      50.71
Yes     49.29
Name: proportion, dtype: float64
```

StreamingMovies:

StreamingMovies is a binary categorical variable representing customers who opted for streaming movie services through the company. The dataset shows most customers do not subscribe to streaming movie services at 51.1% while those who opted for it are at 48.9%. This variable is relevant to Monthlycharge because customers who do subscribe to additional entertainment typically incur higher monthly cost.

```
StreamingMovies
No      5110
Yes     4890
Name: count, dtype: int64
StreamingMovies
No      51.1
Yes     48.9
Name: proportion, dtype: float64
```

TechSupport:

TechSupport is a binary categorical variable representing customers who opted for technical support add-on services through the company. The dataset shows most customers do not subscribe to technical support services at 62.5% while those who opted for it are at 37.5%. This variable is relevant to Monthlycharge because customers who do subscribe to additional tech support services incur higher monthly charges.

```
TechSupport
No      6250
Yes     3750
Name: count, dtype: int64
TechSupport
No      62.5
Yes     37.5
Name: proportion, dtype: float64
```

OnlineSecurity:

OnlineSecurity is a binary categorical variable representing customers who subscribed to additional online security services provided by the company. The dataset shows 64.24% of customers did not opt for the service while 35.76% customers did opt for the add on. This variable is relevant to MonthlyCharge because customers who opt for additional security features may incur higher monthly cost.

```
OnlineSecurity
No      6424
Yes     3576
Name: count, dtype: int64
OnlineSecurity
No      64.24
Yes     35.76
Name: proportion, dtype: float64
```

Area:

Area is a nominal categorical variable representing the customers area demographic. The column has values of **Suburban**, **Rural** and **Urban**. The dataset shows that most customers are in suburban areas at 33.46%, rural areas at 33.27%, urban areas at 33.27%. This variable is relevant to MonthlyCharge as different area may charge higher or lower due short reach of service.

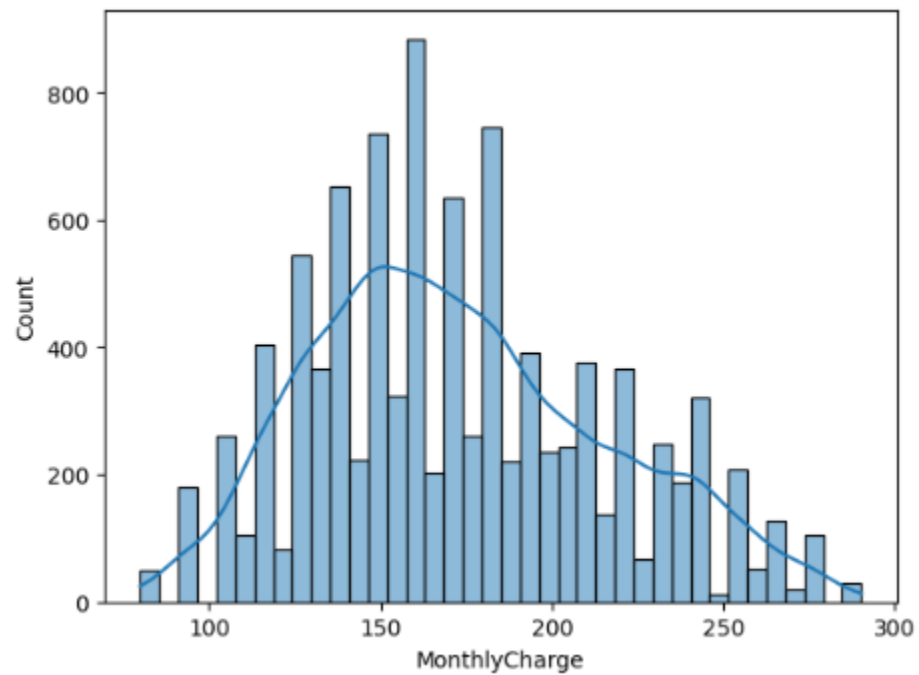
```
Area
Suburban    3346
Rural       3327
Urban       3327
Name: count, dtype: int64
Area
Suburban    33.46
Rural       33.27
Urban       33.27
Name: proportion, dtype: float64
```

Statistical Visualizations:

Univariate Graphs:

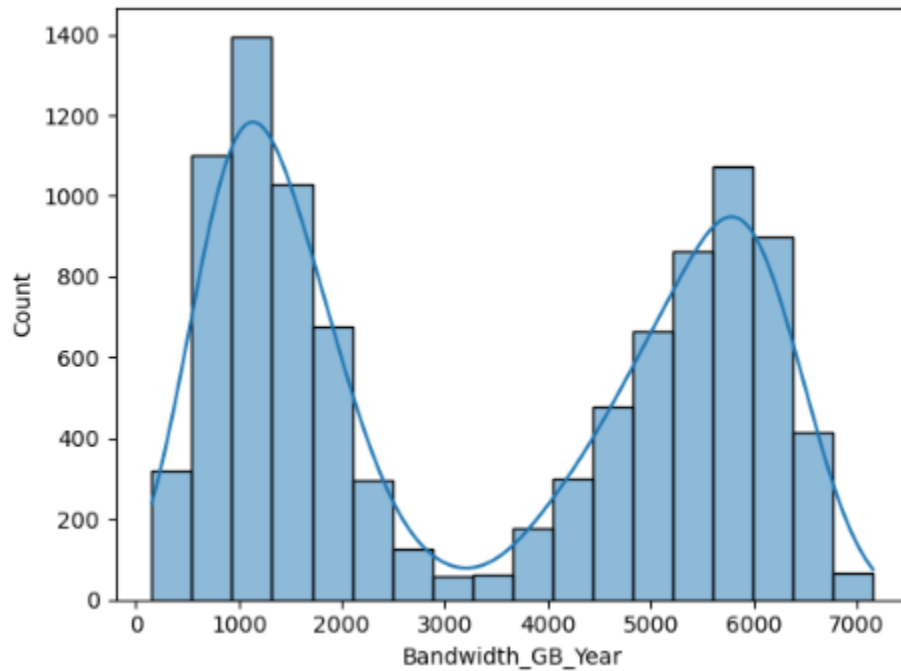
Response Variable Univariate:

MonthlyCharge

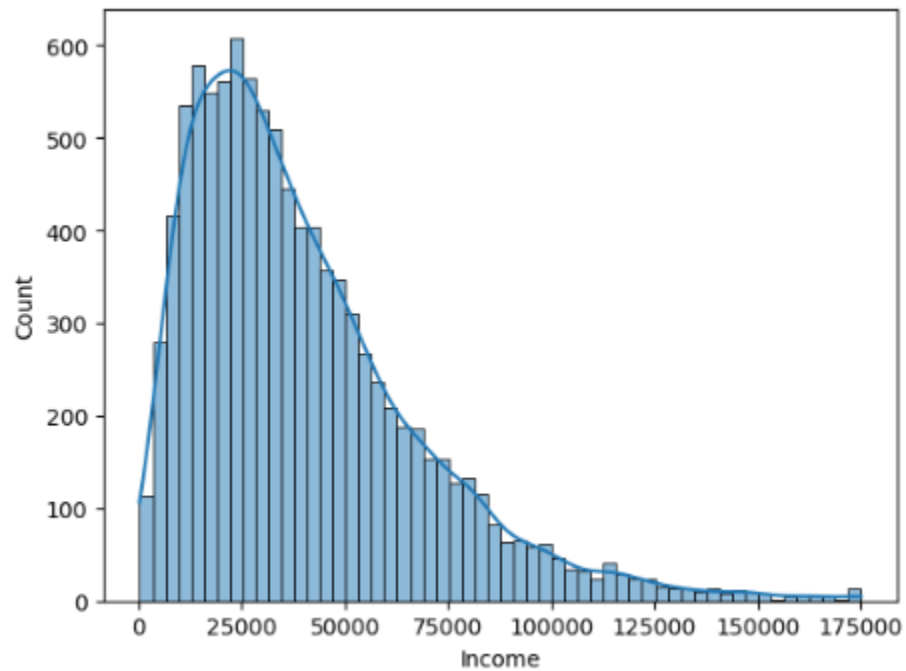


Unimodal Distribution with a slight right skew.

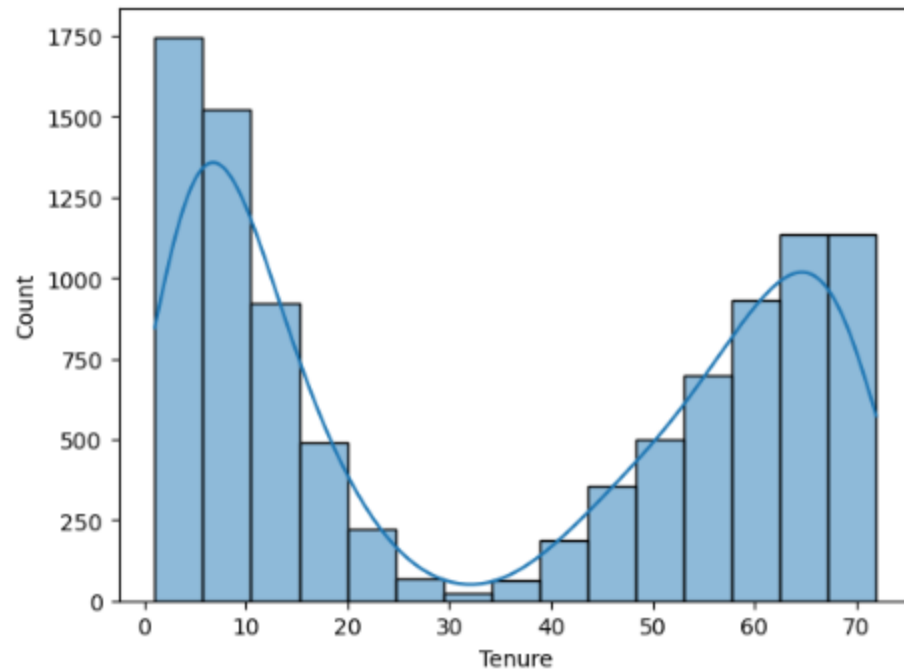
Explanatory Variables Univariate:



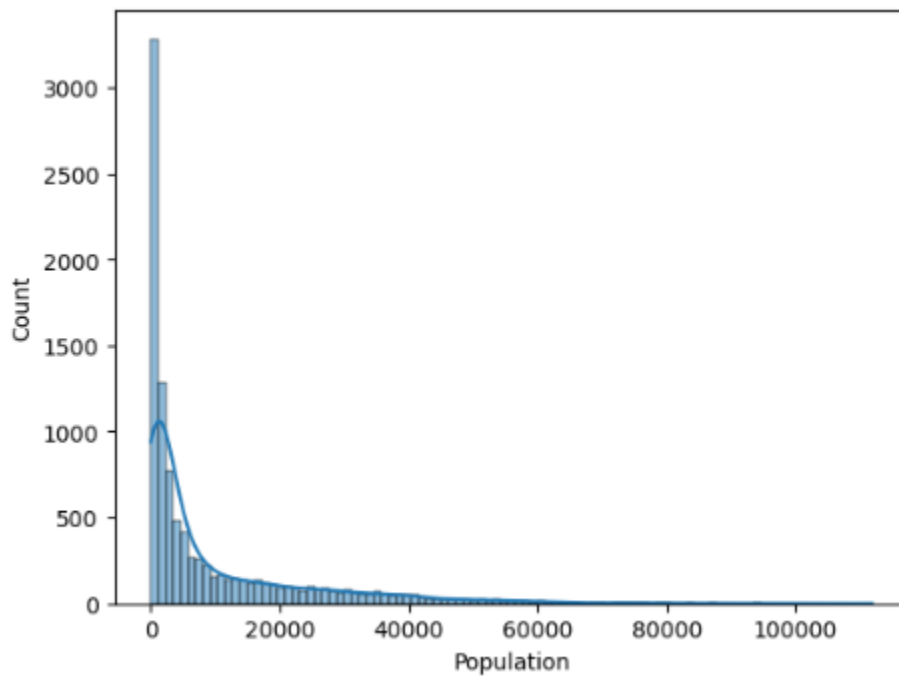
Bimodal distribution with a stronger group on left.



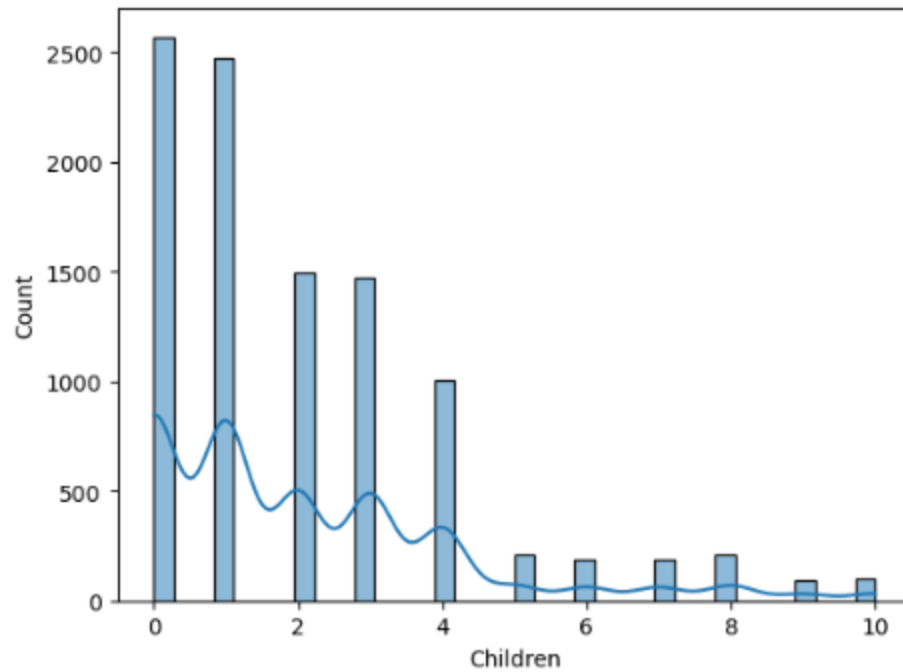
Unimodal distribution with a strong right skew due to outliers of high income earners.



Bimodal distribution showing strong groups of newer and longest tenure customers



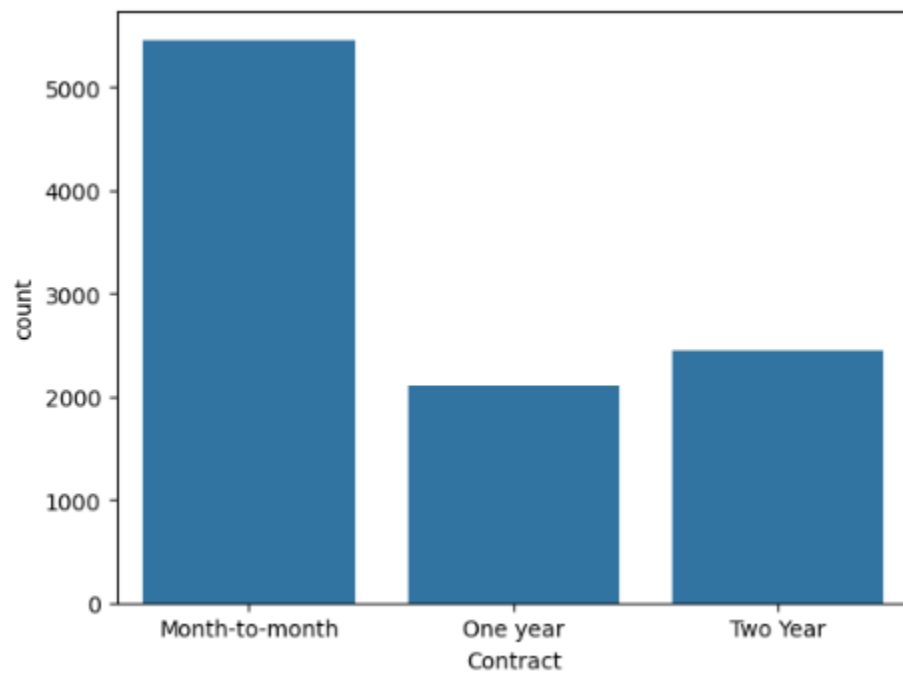
Unimodal Distribution with strong right skew due to urban areas.



Unimodal Distribution with right skew due to some records of having outliers at ten children.

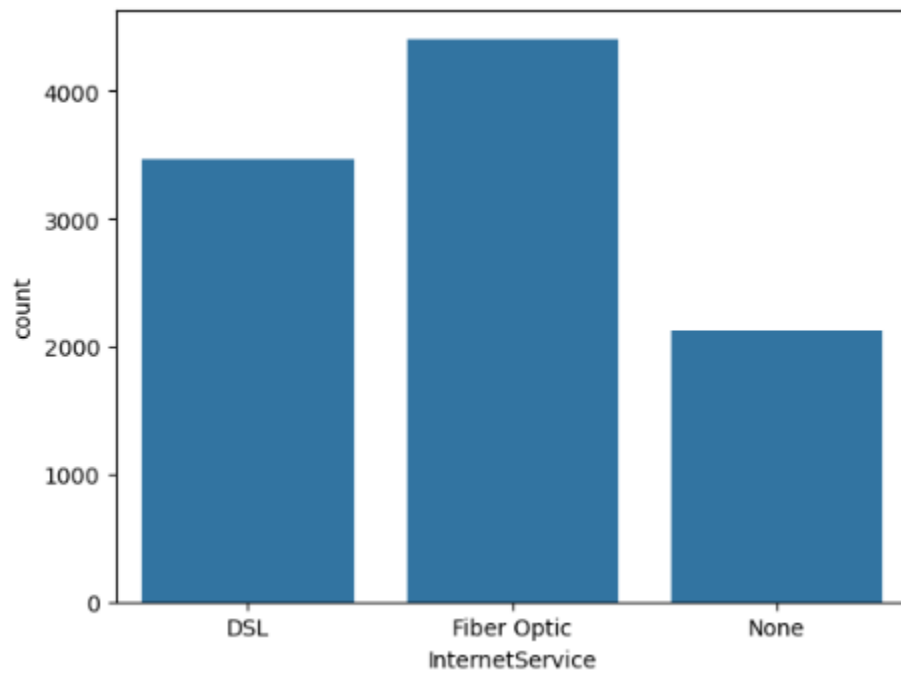
Categoricals Univariate:

Contract:



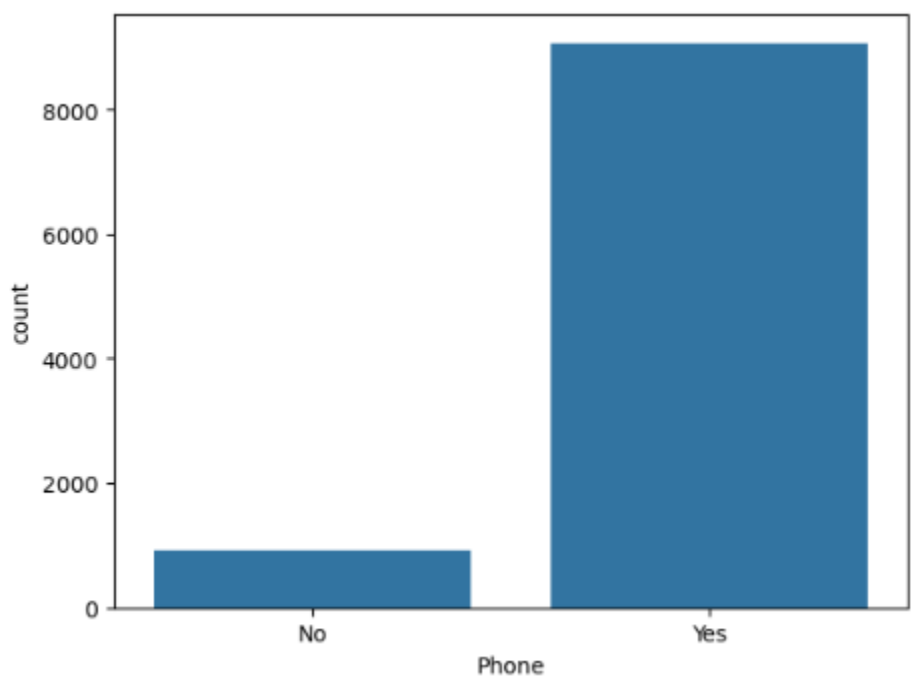
Barplot with Month-to-month Contracts being the most popular.

InternetService:



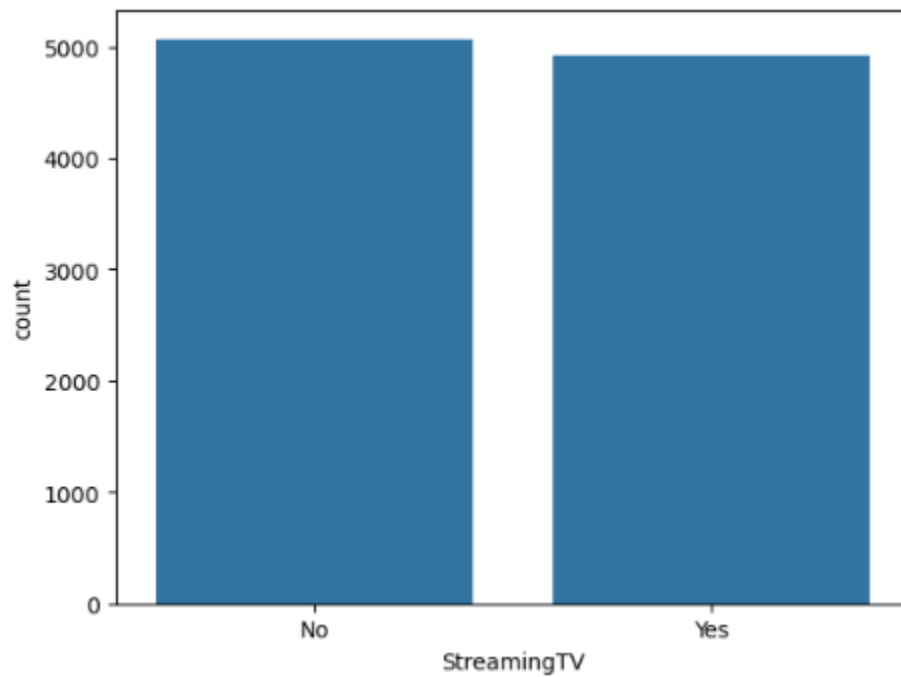
Bar plot of InternetService, with Fiber Optic being the most popular.

Phone:



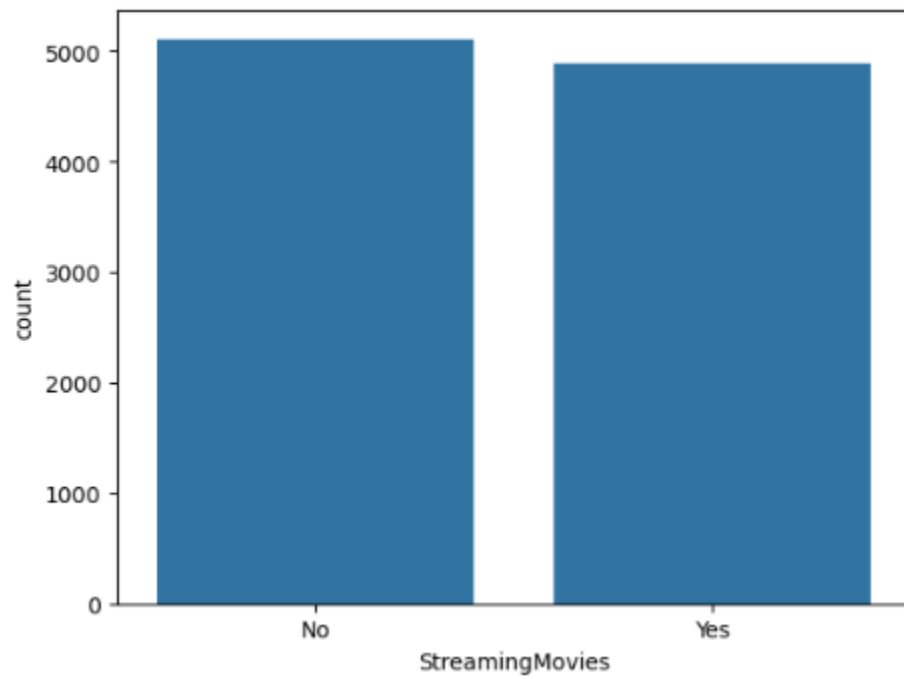
Bar plot of Phone with majority of customers subscribed to phone service.

StreamingTV



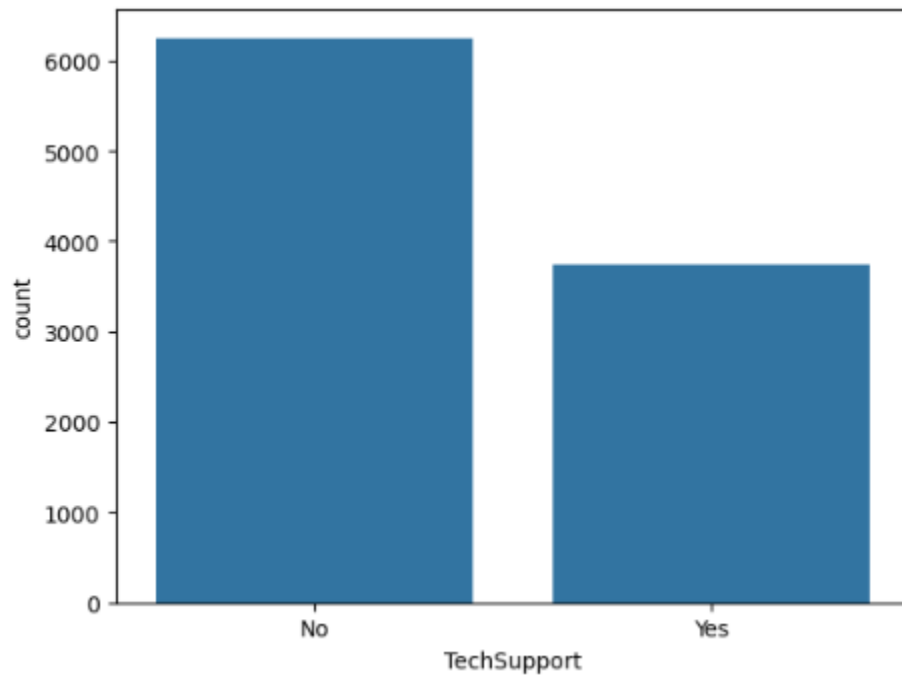
Bar plot of StreamingTV with an almost equal split distribution of those subscribed to TV streaming services through this company.

StreamingMovies:



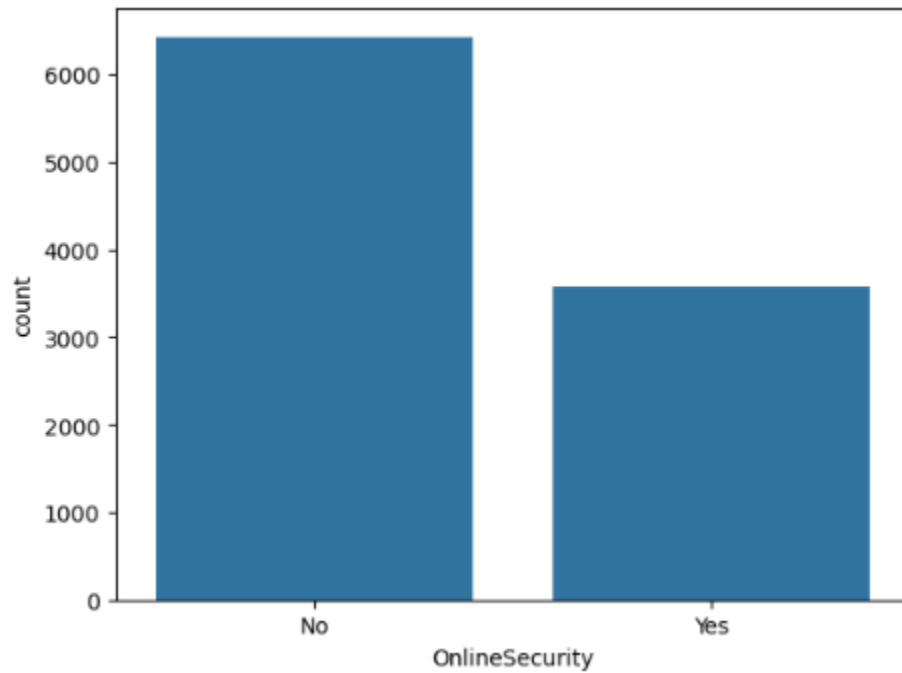
Bar plot of StreamingMovies with a fairly close split distribution of customers who subscribed to streaming movies service through this company.

TechSupport:



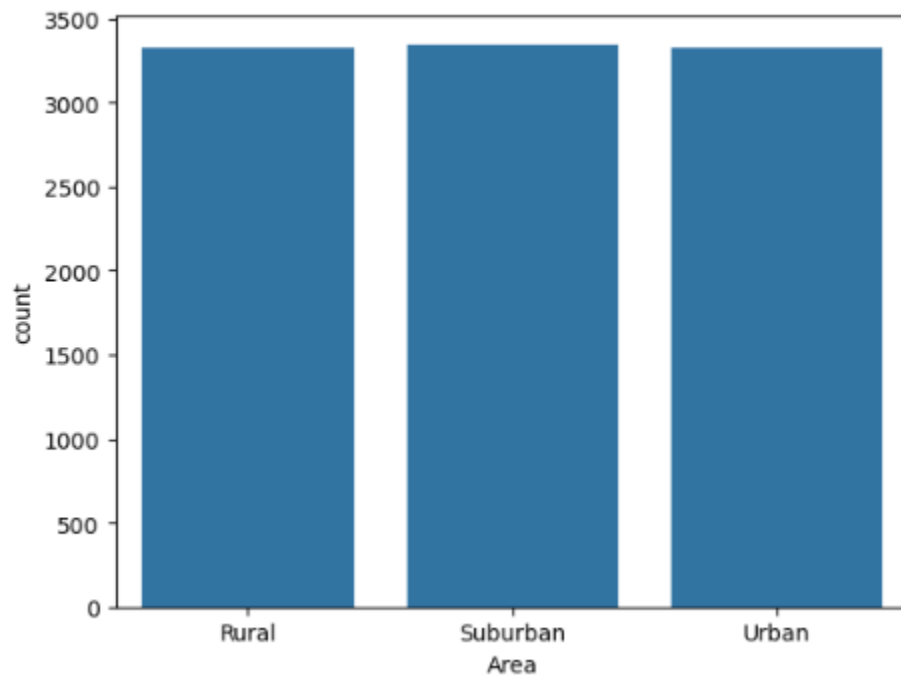
Bar plot of Techsupport, with majority of customers not subscribed to the tech support add on at an almost 60/40 split.

OnlineSecurity:



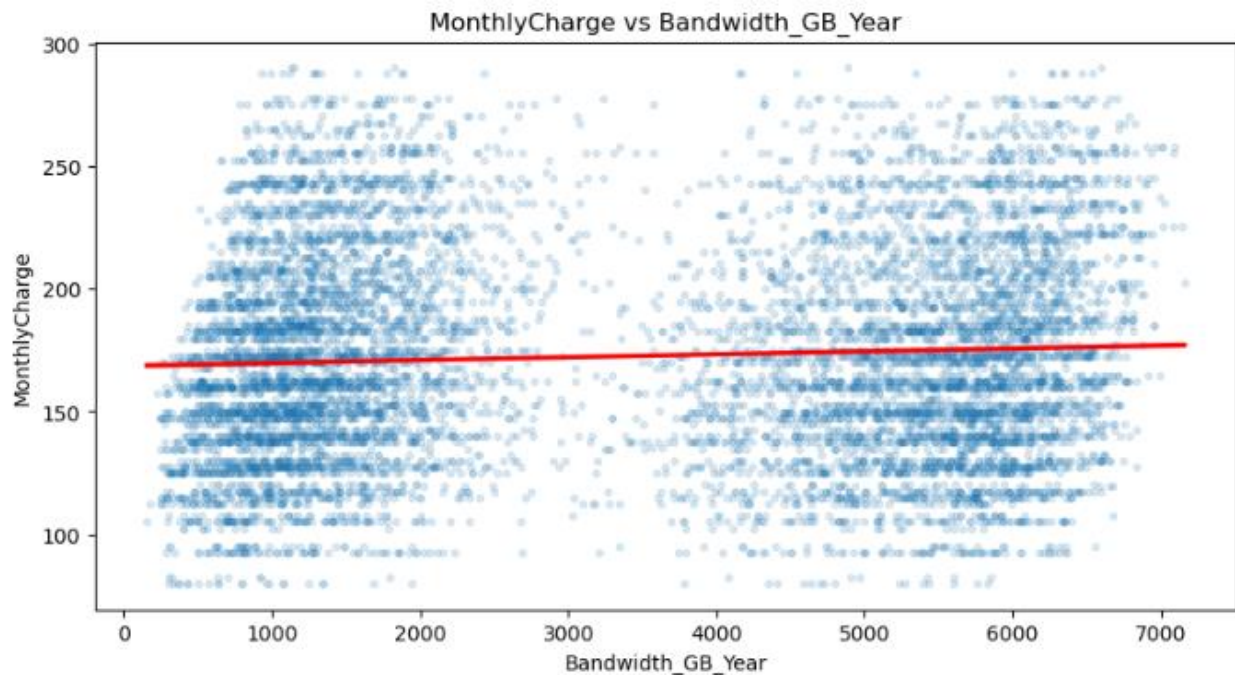
Bar plot of OnlineSecurity, with majority of customers not subscribed to the security add on

Area:

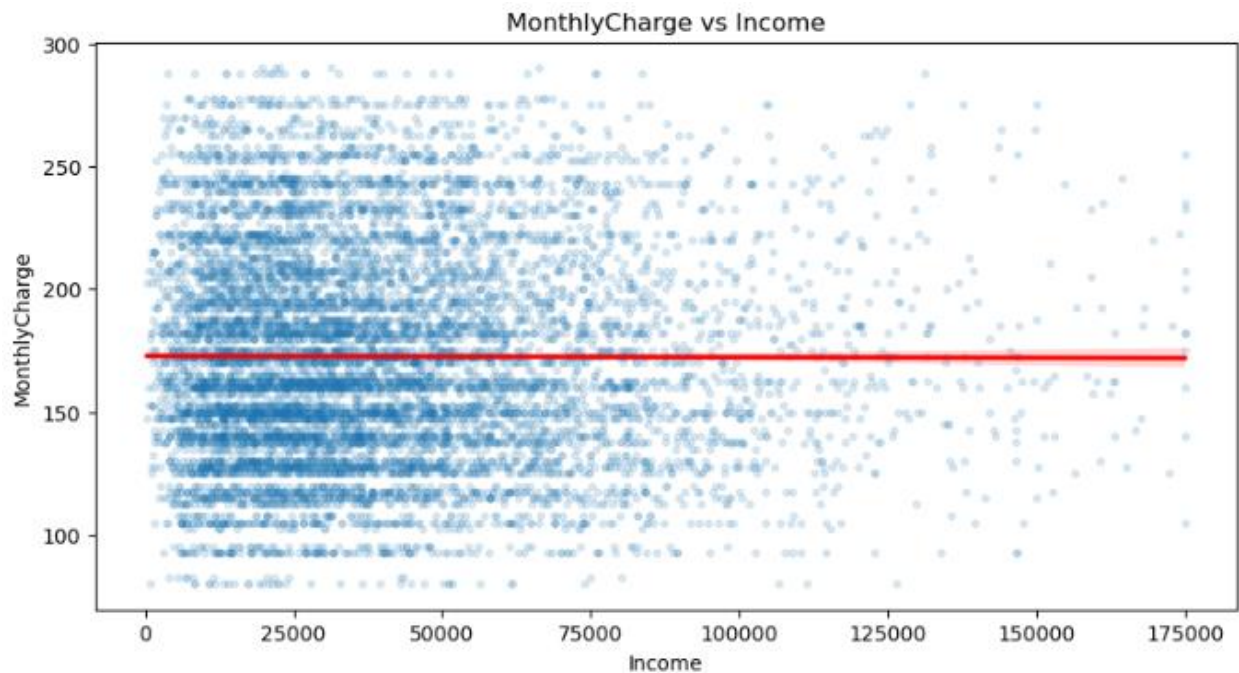


All customer area demographics are evenly distributed through different area demographic.

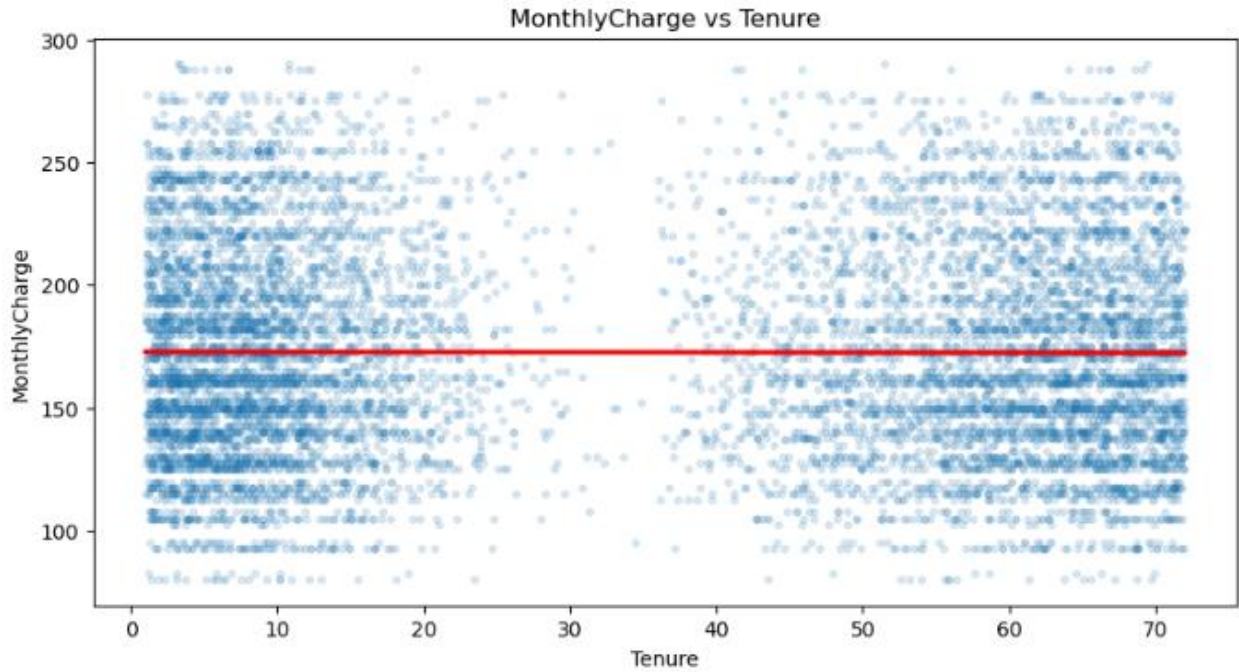
Bivariate Graphs with Response + Explanatory



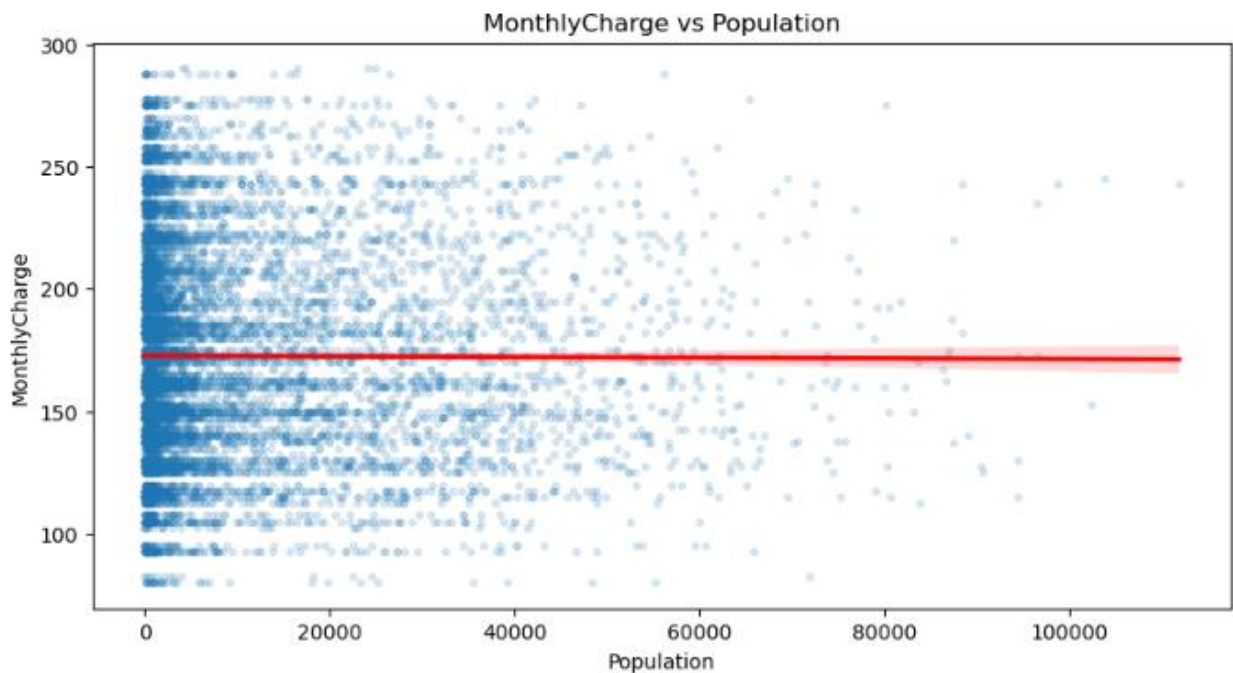
Widespread of MonthlyCharge values at all bandwidth levels, slight upward trend of the fitted line. Two visible clusters.



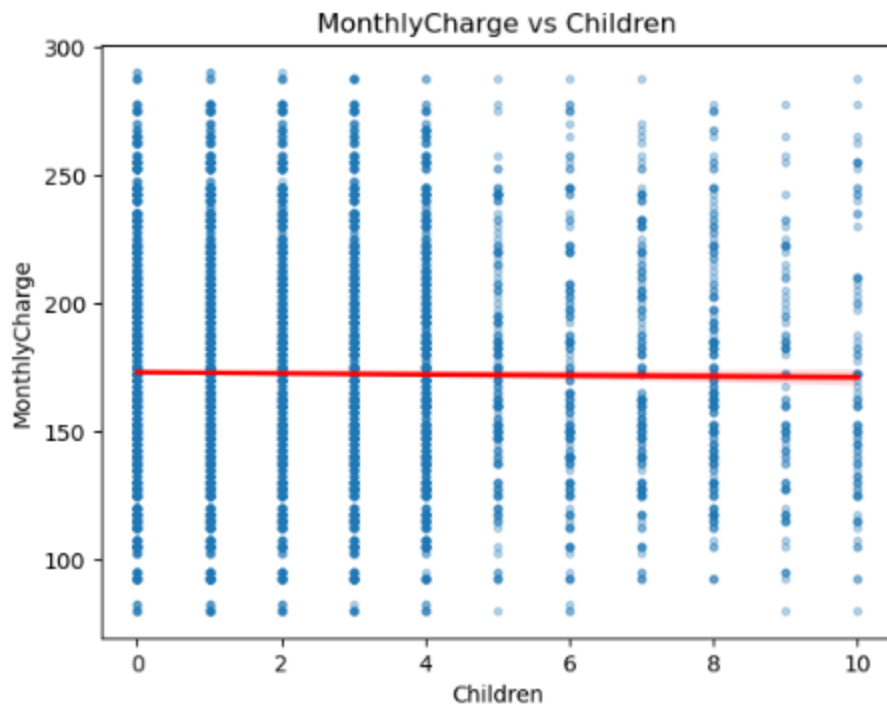
Dense cluster at lower incomes with no strong linear relationship across income.



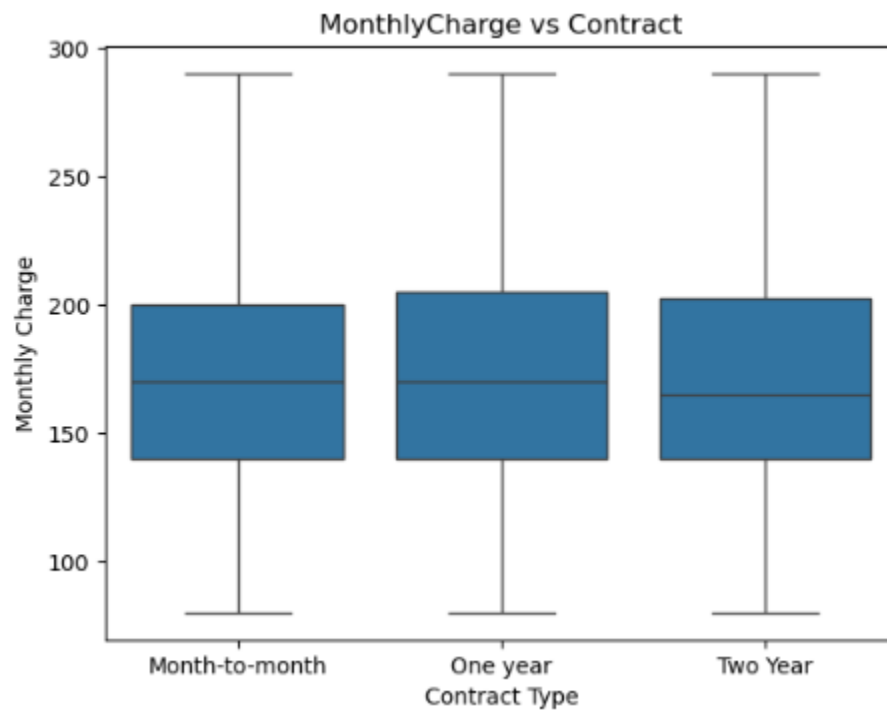
Distribution density is split between long tenures and new customers. No strong linear relationship between MonthlyCharge and Tenure. Wide vertical spread of MonthlyCharge.



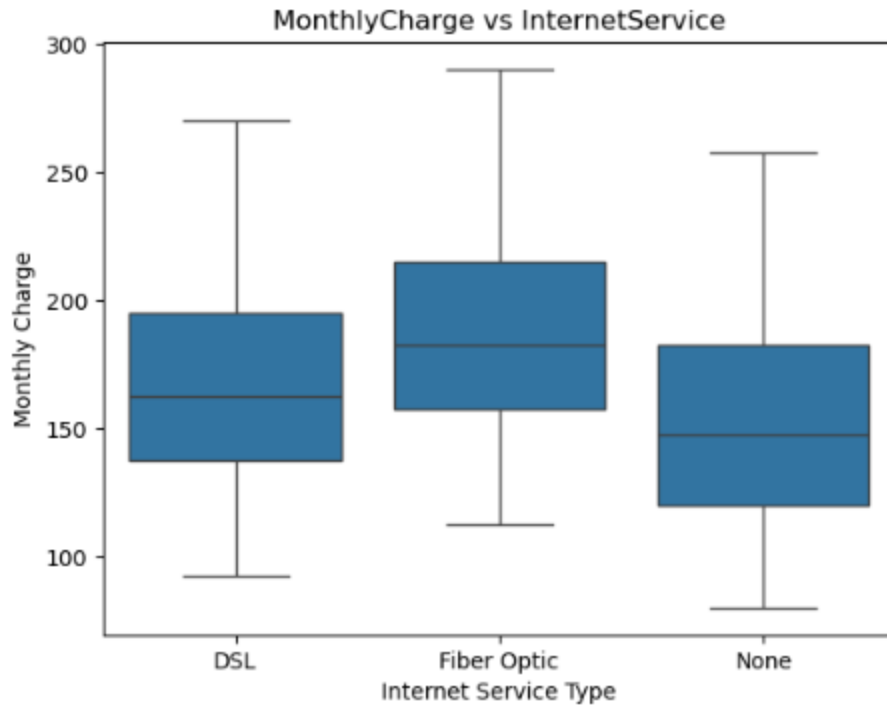
Strong horizontal spread with strong cluster at areas with small population. Vertical spread with strong density in center. Weak linear relationship.



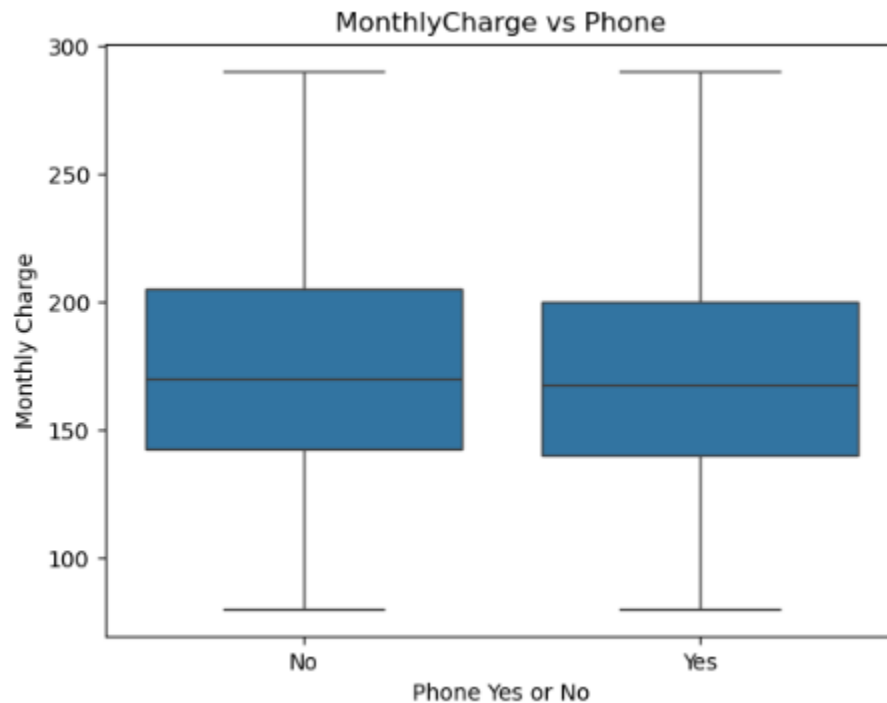
Shows substantial variability with less density after 4 children. Weak linear relationship. MonthlyCharge remains well spread at 4 and less children, with more variability afterwards.



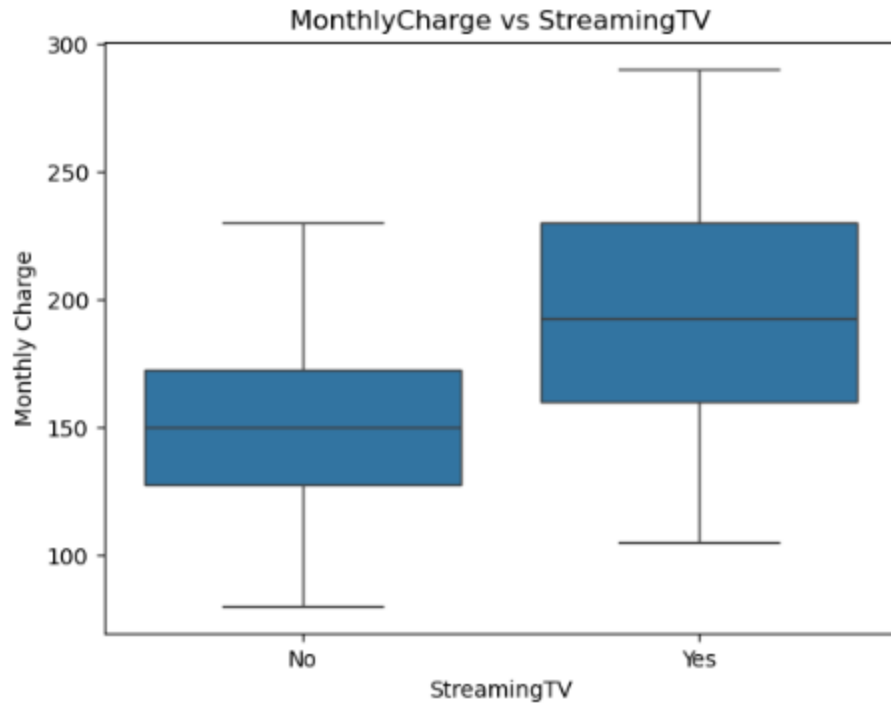
Similar distribution for all values of Contract type.



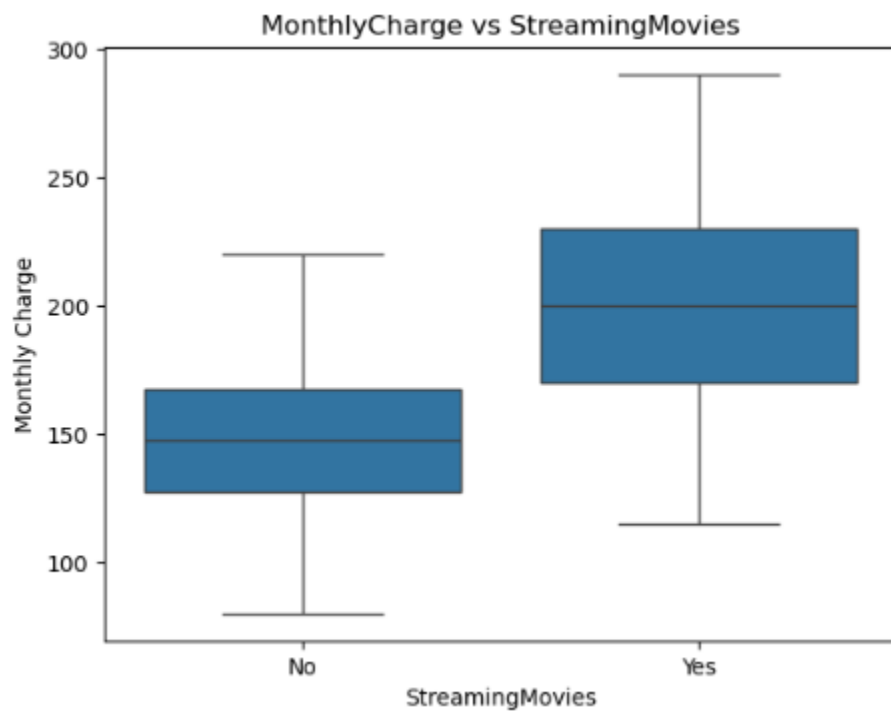
Fiber Optic shows to have a higher initial price, with higher median, while no internet service exhibits the lowest price in all statistics.



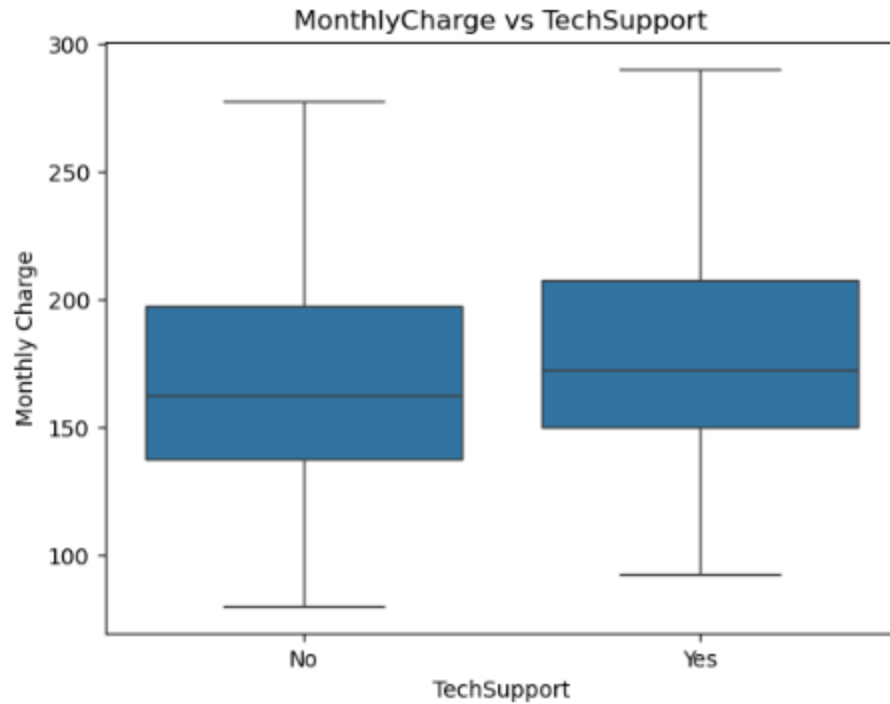
Distribution between those with phone service and none in pricing is very close.



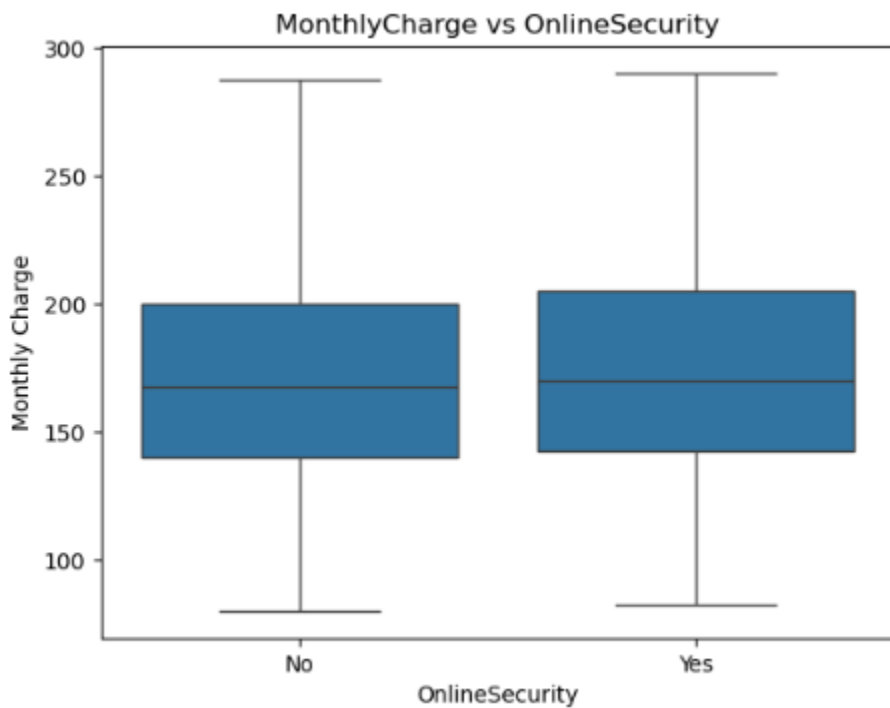
Significantly higher MonthlyCharges for those with streaming tv add on.



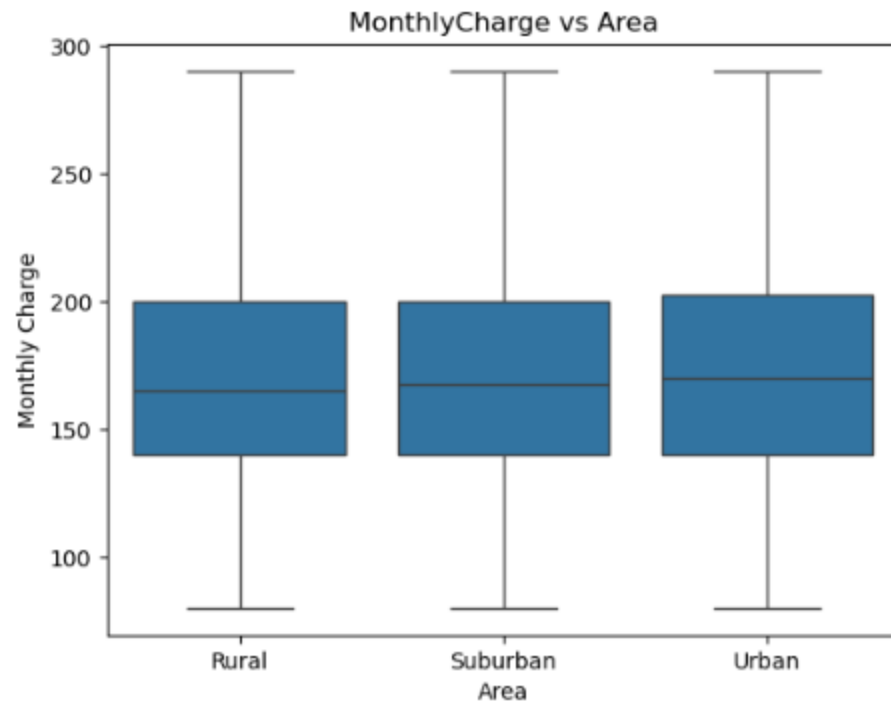
Significantly higher MonthlyCharges for those with streaming movies add on.



Slightly higher monthly charges for those with tech support add on on all statistics.



Small change in monthly charges for those with online security add on, similar median, slightly higher quartile 1 and quartile 3.



Very equal distribution in monthly charges for all demographic.

Data Transformation Process to achieve goal:

The goals of the data transformation were to prepare the dataset for multiple linear regression analysis using **Ordinary Least Squares (OLS)**. Because OLS requires all predictors to be numeric and free of multicollinearity, categorical variables were transformed into numeric representations and one hot encoded in order to be interpretable by the linear regression algorithm.

Several categorical variables in the dataset were binary in nature (Yes/No values). These variables were converted into numeric indicator variables where Yes = 1 and No = 0. This transformation allows binary features to be directly interpreted as difference in MonthlyCharge.

```
# Turn Categorical yes no's into Binary
```

```
df_model['Phone'] = df_model['Phone'].replace({'Yes': 1, 'No': 0})
```

```
df_model['StreamingTV'] = df_model['StreamingTV'].replace({'Yes': 1, 'No': 0})
```

```
df_model['StreamingMovies'] = df_model['StreamingMovies'].replace({'Yes': 1, 'No': 0})
```

```
df_model['TechSupport'] = df_model['TechSupport'].replace({'Yes': 1, 'No': 0})
```

```
df_model['OnlineSecurity'] = df_model['OnlineSecurity'].replace({'Yes': 1, 'No': 0})
```

```
#turn binary categoricals into Int datatype
```

```
into_int = ['Phone','StreamingTV','StreamingMovies','TechSupport','OnlineSecurity']
```

```
for col in into_int:
```

```
    df_model[col] = df_model[col].astype('int')
```

Categorical nominal variables with more than two categories (Area, Contract, InternetService) were converted using one-hot encoding with `getdummies()`. For each variable, one category was dropped to serve as the reference group ($K-1$ = number of columns – 1), preventing multi collinearity/perfect linearity in the regression model.

#One hot encode categorical nominal categories

```
one_hot_category= ['Contract','InternetService','Area']
```

```
df_model=pd.get_dummies(df_model,  
                        columns=one_hot_category,  
                        drop_first=True  
                        )
```

Model Creation Process:

An initial multiple linear regression model was constructed using **OLS** with MonthlyCharge as the dependent variable. All variables identified in C2 are included.

```
#Create first model with all selected variables from C@
```

```
initial_model = ols("MonthlyCharge ~ Bandwidth_GB_Year + Income + Tenure + Population +  
Children + Phone + StreamingTV + StreamingMovies"
```

```
    "+ TechSupport + OnlineSecurity + Contract_One_year + Contract_Two_Year +  
InternetService_Fiber_Optic "
```

```
    "+ InternetService_None + Area_Suburban + Area_Urban", data = df_model).fit()
```

```
#summary
```

```
print(initial_model.summary())
```

OLS Regression Results

```

=====
Dep. Variable:      MonthlyCharge    R-squared:                0.844
Model:              OLS              Adj. R-squared:           0.844
Method:             Least Squares    F-statistic:              3370.
Date:               Tue, 23 Dec 2025  Prob (F-statistic):          0.00
Time:               04:15:00         Log-Likelihood:           -42505.
No. Observations:   10000           AIC:                      8.504e+04
Df Residuals:       9983            BIC:                      8.517e+04
Df Model:           16
Covariance Type:    nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	41.9379	1.185	35.380	0.000	39.614	44.261
Contract_One_year[T.True]	0.6143	0.436	1.408	0.159	-0.241	1.470
Contract_Two_Year[T.True]	0.1363	0.414	0.329	0.742	-0.675	0.948
InternetService_Fiber_Optic[T.True]	76.9616	0.761	101.126	0.000	75.470	78.453
InternetService_None[T.True]	43.5999	0.809	53.917	0.000	42.015	45.185
Area_Suburban[T.True]	0.2113	0.416	0.508	0.612	-0.604	1.027
Area_Urban[T.True]	0.4215	0.417	1.012	0.312	-0.395	1.238
Bandwidth_GB_Year	0.1375	0.002	86.958	0.000	0.134	0.141
Income	4.453e-06	6.07e-06	0.733	0.464	-7.45e-06	1.64e-05
Tenure	-11.2580	0.130	-86.892	0.000	-11.512	-11.004
Population	1.11e-05	1.18e-05	0.942	0.346	-1.2e-05	3.42e-05
Children	-4.4785	0.094	-47.884	0.000	-4.662	-4.295
Phone	0.0143	0.584	0.025	0.980	-1.131	1.160
StreamingTV	10.6891	0.496	21.542	0.000	9.716	11.662
StreamingMovies	23.6623	0.477	49.625	0.000	22.728	24.597
TechSupport	12.1045	0.351	34.471	0.000	11.416	12.793
OnlineSecurity	-8.0756	0.377	-21.432	0.000	-8.814	-7.337

```

=====
Omnibus:            239.590    Durbin-Watson:           2.020
Prob(Omnibus):      0.000    Jarque-Bera (JB):        125.595
Skew:               0.054    Prob(JB):                 5.34e-28
Kurtosis:           2.462    Cond. No.                  4.50e+05
=====

```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 4.5e+05. This might indicate that there are strong multicollinearity or other numerical problems.

Justification for model reduction:

The reducing of the initial multiple linear regression model, **backward stepwise elimination** is used.

This method requires an initial model with all candidate variables and iteratively removes predictor (X) variables that are not statistically significant based on their P-Values as to reduce noise. A significance level of 0.05 was used as the threshold for candidates of removal on the variable. Variables with p-values greater than 0.05 fail to show sufficient significance of having a linear relationship with the response variable (MonthlyCharge).

After each removal of a predictor variable, model performance was evaluated using R-Squared + Adjusted R-squared + prob(F-statistic). The R square and adjusted R-squared are prioritized as it explains the variation in MonthlyCharge. Adjusted R-squared is prioritized as it does not increase when adding new variables, therefore it keeps the evaluation clean as it only increases when the variables add to the model while R-squared increases with more predictors. Therefore, adjusted R-squared is more prioritized than R-squared. Prob(F-Statistic) was used to confirm that the overall reduced model remained statistically significant as a whole. No decrease in adjusted R-squared and no increases in Prob(F-Statistic) justifies the elimination of the weak candidate variables.

This procedure aligns with the research objective of identifying meaningful drivers of MonthlyCharge as Backwards Stepwise elimination can remove weak variables while maintaining a high R-Squared/Adj R-squared metric and low Prob(F-Statistic) to ensure a model with strong contributing variables. In a separate model meant for experimentation, important contributing variables such as Children were removed from the model, showing noticeable decline changes in Adjusted R-squared, confirming their useful contribution to the model and validating their retention in the final model.

```
#Create first model with all selected variables from C2
```

"+ TechSupport + OnlineSecurity + Contract_One_year + Contract_Two_Year + InternetService_Fiber_Optic "

#summary

```
print(initial_model.summary())
```

OLS Regression Results						
=====						
Dep. Variable:	MonthlyCharge	R-squared:	0.844			
Model:	OLS	Adj. R-squared:	0.844			
Method:	Least Squares	F-statistic:	3370.			
Date:	Fri, 26 Dec 2025	Prob (F-statistic):	0.00			
Time:	08:29:08	Log-Likelihood:	-42505.			
No. Observations:	10000	AIC:	8.504e+04			
Df Residuals:	9983	BIC:	8.517e+04			
Df Model:	16					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	41.9379	1.185	35.380	0.000	39.614	44.261
Contract_One_year[T.True]	0.6143	0.436	1.408	0.159	-0.241	1.470
Contract_Two_Year[T.True]	0.1363	0.414	0.329	0.742	-0.675	0.948
InternetService_Fiber_Optic[T.True]	76.9616	0.761	101.126	0.000	75.470	78.453
InternetService_None[T.True]	43.5999	0.809	53.917	0.000	42.015	45.185
Area_Suburban[T.True]	0.2113	0.416	0.508	0.612	-0.604	1.027
Area_Urban[T.True]	0.4215	0.417	1.012	0.312	-0.395	1.238
Bandwidth_GB_Year	0.1375	0.002	86.958	0.000	0.134	0.141
Income	4.453e-06	6.07e-06	0.733	0.464	-7.45e-06	1.64e-05
Tenure	-11.2580	0.130	-86.892	0.000	-11.512	-11.004
Population	1.11e-05	1.18e-05	0.942	0.346	-1.2e-05	3.42e-05
Children	-4.4785	0.094	-47.884	0.000	-4.662	-4.295
Phone	0.0143	0.584	0.025	0.980	-1.131	1.160
StreamingTV	10.6891	0.496	21.542	0.000	9.716	11.662
StreamingMovies	23.6623	0.477	49.625	0.000	22.728	24.597
TechSupport	12.1045	0.351	34.471	0.000	11.416	12.793
OnlineSecurity	-8.0756	0.377	-21.432	0.000	-8.814	-7.337
=====						
Omnibus:	239.590	Durbin-Watson:	2.020			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	125.595			
Skew:	0.054	Prob(JB):	5.34e-28			
Kurtosis:	2.462	Cond. No.	4.50e+05			
=====						

Final Model:

#removing Population for backward stepwise elimination

```
model6 = ols("MonthlyCharge ~ Bandwidth_GB_Year + Tenure + Children + StreamingTV + StreamingMovies")
```

```
" + TechSupport + OnlineSecurity + InternetService_Fiber_Optic +  
InternetService_None", data = df_model).fit()
```

```

OLS Regression Results

=====
Dep. Variable:      MonthlyCharge    R-squared:                0.844
Model:              OLS             Adj. R-squared:            0.844
Method:             Least Squares    F-statistic:              5992.
Date:               Fri, 26 Dec 2025  Prob (F-statistic):        0.00
Time:               06:29:55         Log-Likelihood:           -42507.
No. Observations:   10000           AIC:                      8.503e+04
Df Residuals:       9990            BIC:                      8.511e+04
Df Model:            9
Covariance Type:    nonrobust

=====
                    coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept              42.6013      0.973      43.797      0.000      40.695      44.508
InternetService_Fiber_Optic[T.True]  76.9600      0.761     101.185      0.000      75.469      78.451
InternetService_None[T.True]         43.6086      0.808      53.952      0.000      42.024      45.193
Bandwidth_GB_Year         0.1375      0.002      87.029      0.000         0.134         0.141
Tenure                -11.2604      0.129     -86.964      0.000     -11.514     -11.007
Children              -4.4781      0.093     -47.897      0.000         -4.661         -4.295
StreamingTV           10.6881      0.496      21.547      0.000         9.716      11.660
StreamingMovies       23.6528      0.477      49.627      0.000      22.719      24.587
TechSupport           12.1064      0.351      34.489      0.000      11.418      12.794
OnlineSecurity        -8.0762      0.377     -21.445      0.000         -8.814         -7.338

=====
Omnibus:             241.178    Durbin-Watson:           2.020
Prob(Omnibus):        0.000    Jarque-Bera (JB):        126.361
Skew:                 0.056    Prob(JB):                3.64e-28
Kurtosis:             2.461    Cond. No.                3.48e+04

=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.48e+04. This might indicate that there are
strong multicollinearity or other numerical problems.

```

Comparison of initial and final model:

The data analysis process began with an initial multiple linear regression model that included all independent variables identified in question C2. This full model served as the baseline for evaluating the contribution of each predictor in explaining variation in the response variable, **MonthlyCharge**.

A backward stepwise elimination process was applied, as justified in question D2. This approach iteratively removed predictor variables that were not statistically significant in explaining variation in **MonthlyCharge**, as indicated by p-values greater than 0.05. A significance level of 0.05, corresponding to a 95% confidence level, was used as the threshold for variable removal.

The primary model evaluation metrics used to compare iterations of the model were **R-squared, Adjusted R-squared, and Prob(F-statistic)**. Adjusted R-squared was prioritized because it accounts for the number of variables in the model and penalizes the inclusion of weak or irrelevant predictors, making it a more reliable metric for model comparison. Prob(F-statistic) was used to confirm that the overall model remained statistically significant after each variable removal.

Predictor variables with p-values exceeding 0.05 were removed sequentially, including **Phone, Contract, Area, Income, and Population**. After each removal, the evaluation metrics remained unchanged (R-squared = 0.844, Adjusted R-squared = 0.844, Prob(F-statistic) = 0.00). These results indicate that the removed variables did not contribute statistically significant explanatory power to the model and primarily introduced noise rather than predictive value.

Output head of model predictions (First five):

```
: #E2 Professors Middleton webinar ask for predicted values of Y
#Get predicted values of Y, show the first 5 lines
predicted_y = model6.fittedvalues
print(predicted_y.head())

0    182.993996
1    238.460698
2    153.471996
3    145.961967
4    160.871131
dtype: float64
```

Reduced models RSE:

#Calculating reduced models RSE

```
rse= np.sqrt(model6.mse_resid)

print(rse)

print("Residual Standard Error of Reduced Model:",rse)
```

```
#Calculating reduced models RSE

rse= np.sqrt(model6.mse_resid)
print(rse)

print("Residual Standard Error of Reduced Model:",rse)

16.984499693137618
Residual Standard Error of Reduced Model: 16.984499693137618
```

Residual plot:

```
#Residual plot

fitted_vals = model6.fittedvalues

resid = model6.resid

plt.figure(figsize=(8,5))

plt.scatter(fitted_vals, resid, alpha=0.5)

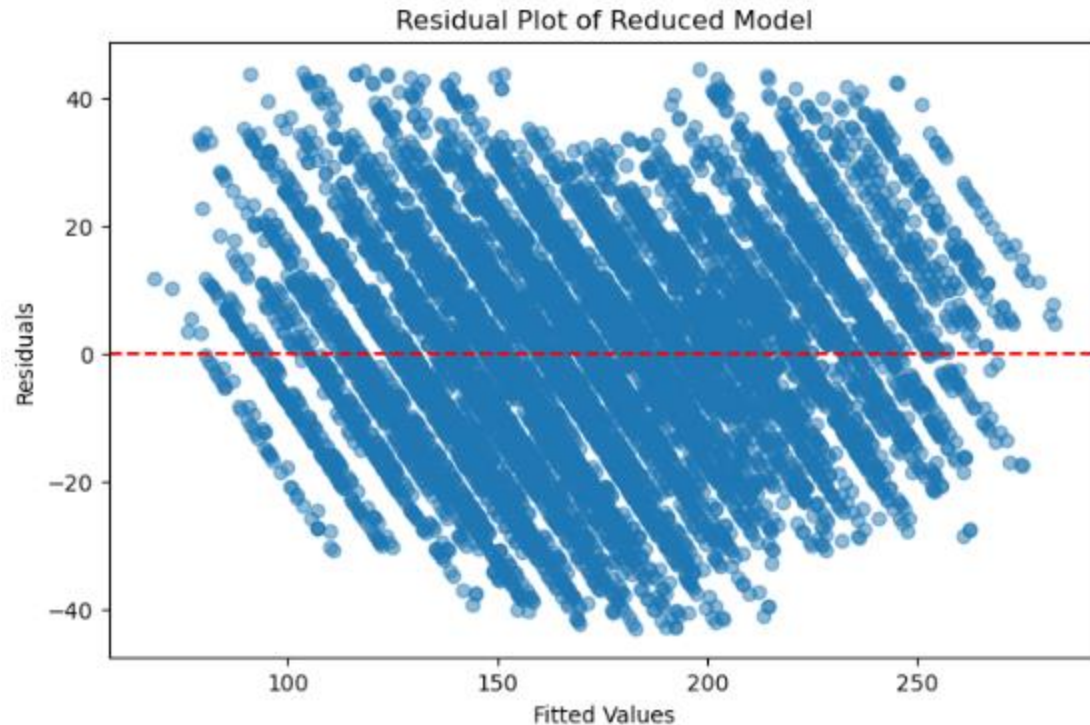
plt.axhline(y=0, color='red', linestyle='--')

plt.xlabel("Fitted Values")

plt.ylabel("Residuals")

plt.title("Residual Plot of Reduced Model")

plt.show()
```



F. Summarize your findings and assumptions by doing the following in a document file:

1. Discuss the results of your data analysis, including the following elements:

- a regression equation for the reduced model
- an interpretation of the coefficients of the reduced model
- the statistical and practical significance of the reduced model
- the limitations of the data analysis

Regression Equation of Reduced model:

MonthlyCharge = 42.6013 + 10.6881(StreamingTV) + 23.6528(StreamingMovies) + 12.1064(TechSupport) – 8.0762(OnlineSecurity) +76.96(InternetService = Fiber) + 43.6086(InternetService = None) + 0.1375(Bandwidth_GB_Year) – 11.2604(Tenure) – 4.4781(Children)

Interpretation of the coefficients of the reduced model:

Intercept(42.6013) This represents the estimated baseline MonthlyCharge for a customer in the reference categories (No service media(TV/Movies) add ons, no tech support, no online security, DSL Internet, no additional bandwidth, zero tenure, and zero children. While this scenario is not realistic, the intercept serves as a baseline for other variables to add on to or subtract.

StreamingTV(10.6881), while holding all other variables constant, customers who subscribed to streaming tv add-ons pay approximately \$10.69 more per month than customers who do not.

StreamingMovies(23.6528) Customers with StreamingMovies service pay on average, \$23.65 more per month.

TechSupport(12.1064) Customers with the TechSupport add-on pay on average \$12.11 more in Monthlycharge.

OnlineSecurity(-8.0762) Customers with the OnlineSecurity add on are associated to have \$8.08 less in Monthlycharges.

InternetService = Fiber(76.96) Customers with fiber internet service on average pay \$76.96 more compared to the baseline internet service (DSL)

InternetService = None(43.6086) Customers with no internet service still have a higher Monthlycharge relative to the baseline category, likely due to other subscribed services

Bandwidth_GB_Year(0.1375) For every additional gigabyte the customer uses increases MonthlyCharge by approximately \$0.14, indicating a small but positive usage-based effect.

Tenure(-11.2604) Each additional unit of tenure(Month) is associated with an average of \$11.26 decrease in MonthlyCharge, suggesting longer term customers receive loyalty discounts.

Children(-4.4781) Each additional child in a household is associated with a decrease of approximately \$4.48, possibly reflecting a family plan.

Statistical and practical significance of the reduced model:

The reduced linear regression model is statistically significant, as indicated by prob(F-Statistic) of 0.00, demonstrating that the retained predictors explain the variation in MonthlyCharge.

The model archived an adjusted R-squared value of 0.844, meaning that approximately 84.4% of the variation in Monthlycharge is explained by the reduced model after accounting for the

number of predictors. The stability of adjusted R-squared across iterations of variable removal indicates that eliminated variables did not meaningfully contribute to explanatory power.

From a practical perspective, several predictors show substantial real world impact. For example, Fiber Optic internet service is associated with an average \$76.96 increase per month, while StreamingTV and StreamingMovies add \$10.69 and \$23.65 respectively. These magnitudes represent meaningful cost differences for customers and are operationally significant for pricing and service strategy decisions.

Limitations of data analysis:

This analysis has several limitations. First, the linear regression model assumes linear relationship between the response and predictor variables, which could oversimplify complex pricing structures as it could be affected by multitudes of reasons.

The model does not include potentially important variables such as limited time promotional discounts, geographic pricing differences such as states due to high cardinality, customer behavior on usage such as customers with StreamingTV/Movies could be the ones with higher bandwidth usage.

The presence of multicollinearity could skew and amplify results, using correlation on the numerical columns of the reduced dataset, Bandwidth_GB_Year and Tenure are highly correlated, indicating overlapping explanatory information. This overlap can amplify the effects of these variables, making it difficult to isolate their individual impact on MonthlyCharge.

Lastly, the analysis identifies associations rather than causal relationships; therefore, conclusions should be interpreted as correlational rather than causation.

Recommendation:

Based on the results of the reduced linear regression model, it is recommended that the company focuses pricing strategy and revenue optimization efforts on service-based add-ons rather than demographic or geographic factors. Variables such as StreamingTV, StreamingMovies, TechSupport and InternetService types were shown to have strong positive associations with Monthlycharge, indicating these features are key drivers of customer billing.

To contrast, demographic variables such as Income, Population, Area and Contract type were not statistically significant enough and were removed during model reduction, suggesting these do not meaningfully contribute to pricing variation. As a result, pricing or marketing strategies based on these attributes may provide limited value.

The negative association between Tenure and MonthlyCharge suggest that long-term customers receive loyalty-based pricing, the company may consider reviewing this to balance customer loyalty incentives for revenue optimization.